

# Force generation in kinesin hinges on cover-neck bundle formation

Wonmuk Hwang, Matthew J. Lang, and  
Martin Karplus

## Supplemental Texts and Figures

### S.1 Choice of solvation models

The neck pulling simulation of 2KIN was performed using the GBMV II continuum solvent model (Lee et al., 2002). However, simulations with GBMV II are considerably slower than those in another continuum solvent, ACE2 (Schaefer and Karplus, 1996; Schaefer et al., 2001), which has been shown to work well for proteins. As ADP turned out not to be directly relevant to the conformational behavior of CNB, we deleted ADP, and used ACE2 and the param19 force field for other simulations. For verification purpose, we also performed explicit water simulations with the param22 force field, in certain cases (as explained below), and found no major difference in the overall behavior except for the increased time scale due to the slow relaxation of explicit water.

### S.2 Pulling the neck helix

2KIN does not have coordinates for the microtubule binding L11 loop. For simulation, they were modeled by using the ModLoop server<sup>1</sup> (Fiser et al., 2000). L11 may affect the rotation of the switch II, which may subsequently initiate CNB formation (Figure 6A). As explained in the text, however, the conformation of L11 itself has no effect on the behavior of CNB.

Different pulling directions for the neck helix (Figure 1B) were used to explore the direction dependence of the unbinding of the neck linker. The pulling directions were calculated based on the following two unit vectors. First, the axis ( $\hat{u}_1$ ) of the neck helix (AA 339-347) was calculated using the COOR HELIX command in CHARMM. The second unit vector  $\hat{u}_2$  was defined in the direction from N334 C $\alpha$  to T338 C $\alpha$  (Figure S8), pointing approximately in the direction of  $\beta 10$ . The 5 pulling directions in Figure 1B are:

$$\begin{aligned}\text{Axis} &: \hat{u}_1 \\ \text{Backward} &: 0.5(\hat{u}_1 - \hat{u}_2) \\ \text{Forward} &: 0.5(\hat{u}_1 + \hat{u}_2) \\ \text{Left} &: 0.5(\hat{u}_1 + \hat{u}_1 \times \hat{u}_2) \\ \text{Right} &: 0.5(\hat{u}_1 - \hat{u}_1 \times \hat{u}_2).\end{aligned}$$

After converting 2KIN from raw PDB to the param22 PDB format, a short initial minimization was performed to remove bad contacts. For MD, the Nosé-Hoover thermostat (Hoover, 1985) was used with a 2 fs integration time step (unless noted, this integration scheme was used for

other simulations as well). The system was heated from 98 K to 300 K for 3 ps. Harmonic constraints were then applied to constrain C $\alpha$  atoms in the microtubule binding domains (AA 142-145 in  $\beta 5$ , AA 273-281 in the switch II, and AA 238 and 255 in L11; Figure S7). Each constraint had a spring constant of 10 kcal/(mol·Å<sup>2</sup>). Pulling was performed at 300 K for 400 or 800 ps, depending on when the neck unbinds. It was also performed on 1MKJ, but in the ACE2 continuum solvent. This was mainly to prepare an unbound conformation of 1MKJ that was later used for other simulations.

### S.3 Motion of unbound head and re-binding of CNB

Initial structures in Figure 2D,E were used for simulations of freely moving motor heads. They were prepared from the coordinate trajectories of the neck pulling simulations in the axial direction (Figure 1B). To avoid too much deformation in the initial structure, **structures right after unbinding of  $\beta 9$  were taken** rather than those at the end of the simulations. In simulations without the cover strand, AA 1-8 was cleaved.

Initial **energy minimization was performed** for a total of 2000 minimization steps (400 steps using the steepest descent method, followed by 1600 steps using the adopted basis Newton-Raphson method. Similar minimization schemes are used for other cases below). As a reference for the motion of the unbound motor head, harmonic constraints were applied to fix the C $\alpha$  atoms of the neck helix that had been used to pull during the neck pulling simulation. Production runs were performed at 360 K for 4-8 ns (Figure 2B,C and Figure S10). The higher temperature was used to speed up the motion and to allow a larger conformational motion than at 300 K.

The same initial structures as above were used for re-binding simulations (Figure 2D,E), with the same harmonic constraints applied as in the neck pulling simulation. The production run lasted 200-400 ps at 300 K. For the CNB-1BG2 chimera (Figure 2F,G), the following steps were taken: The base of CNB (AA 10-15, 314-320) of the initial structure in Figure 2D was superposed on that in 1BG2, then AA 1-15 and 321-349 were merged to 1BG2 by replacing the the coordinates of the corresponding atoms with those of CNB. The merged region was harmonically constrained and 2400 steps of initial energy minimization was performed. The presence of the extra helical turn of  $\alpha 6$  in the merged region (*cf.*, Figure 6A) displaced the switch II, so that it was positioned similar to the conformation in 1MKJ. The rest of the simulation on the chimera corresponds to those for Figure 2D,E.

As seen in Figure 2F,G, **L13 protrudes and interferes with the the neck linker binding. L13 belongs to the extended switch II cluster** (Figure S7), **so the conformational change upon ATP binding** (Sindelar et al., 2002) **could remove L13 from a protruding position to accommodate the neck linker.** Corresponding conformational

<sup>1</sup><http://alto.compbio.ucsf.edu/modloop/modloop.html>

behavior of L13 is also observed in other PDB structures (Figure S12). Experimentally, the G291A/G292A mutations in L13 led to 100 fold decrease in the microtubule gliding velocity (Case et al., 2000; Wade and Kozielski, 2000). Whereas this was explained by disruption of the N327-G292 hydrogen bond observed in 1MKJ (Case et al., 2000), our contact analysis shows that this bond is between the backbone oxygen of G291 and the non-polar methylene hydrogen of the N327 side chain, which is not a good hydrogen bond donor (not shown in Figure S8D). This contact is also absent in the higher resolution structure of 2KIN (Figure S8C). Instead, we believe that the mutation from G to A reduces the flexibility of L13, sterically blocking tight binding of the neck linker.

#### S.4 Bending of isolated CNBs in explicit water

To verify the local conformational bias of CNB, explicit water simulations were performed on isolated CNBs (Figure S13). We used three different initial configurations, one from the neck pulling simulation of 1MKJ (Figure 2D), and two from the force map simulation (Figure 5). From the neck pulling simulation, AA 1-12 (cover) and 317-333 (neck) were used (Figure S13B), while the structures from the force map simulation contained AA 1-12 and 317-331 (Figure S13C,D). They were first converted from param19 to param22. A box of pre-equilibrated TIP3 water molecules was constructed around the system with a periodic boundary condition in each direction in the Cartesian coordinate system. The size of the periodic box was determined by adding 15 Å to the maximum or minimum coordinates of CNB in each direction. Care was taken to make sure that CNB is at least 15 Å away from the boundary even when it changed conformation. As a 13 Å cutoff of nonbonded interactions was used, boundary effect was negligible. Water molecules whose oxygens lie within 2.2 Å from any of the heavy atoms in CNB were deleted. After solvation, there were up to 7400 water molecules left in the box of size reaching  $53 \times 60 \times 74$  (Å<sup>3</sup>).

To anchor the base of CNB, C<sub>α</sub> atoms of AA 9-12 and AA 317-323 were fixed in space. Conformations of these regions are nearly the same in both ADP (1BG2) and ATP-like (1MKJ) structures. The system was then energy minimized with harmonic constraints applied to the backbone heavy atoms of CNB to eliminate bad contacts. For MD, the standard leap-frog integrator was used instead of the Nosé-Hoover thermostat. Heating was performed from 0 K to 300 K in 10 ps, with the backbone of CNB still constrained. The next step of equilibration maintained the temperature within  $300 \pm 5$  K, and was divided in two stages; the backbone harmonic constraint was present for the first 10 ps, then was removed during the next 10 ps. The production run lasted between 0.6-1 ns.

#### S.5 Molecular force generated by CNB

Let  $E(\vec{r})$  be the free energy of CNB parametrized by the position  $\vec{r}$  of V331 C<sub>α</sub> in 1MKJ. The average molecular force exerted at this location is  $\vec{F}(\vec{r}) = -\nabla E$ . Calculating the force by numerically constructing the free energy landscape and taking the gradient would be a formidable task: To get the free energy landscape, one would need to employ techniques such as umbrella sampling (Kotlam and Case, 1988). This would require scanning through a wide range of  $\vec{r}$ , with overlapping intervals in three dimensions, which would cause the gradient highly noisy unless very long simulation time is achieved. For the purpose of our work, and considering that forces are directly measured in optical trap experiments, it is not necessary to measure the free energy. We instead developed an algorithm that measures the free energy gradient directly from the coordinate trajectory.

Consider a 1-dimensional (1-D) system, whose free energy at a point  $x$  is  $E(x)$ . To calculate the gradient of  $E$  at a particular point  $x_0$ , we add a constraining harmonic potential, so that the overall free energy is

$$\mathcal{E}(x) = E(x) + \frac{k}{2}(x - x_0)^2. \quad (1)$$

The spring constant  $k$  should be large enough so that under the new Hamiltonian, the system fluctuates around the neighborhood of  $x_0$ . We can then Taylor expand  $E(x)$  up to a second order,

$$\begin{aligned} E(x) &= E(x_0) + (x - x_0)E' + \frac{(x - x_0)^2}{2}E'' \\ &+ O(|x - x_0|^3 E'''), \end{aligned} \quad (2)$$

where  $E' = E'|_{x=x_0}$ , and  $E'' = E''|_{x=x_0}$ , etc. Using  $y = x - x_0$ ,

$$\mathcal{E} \simeq E(x_0) + E'y + \frac{1}{2}(k + E'')y^2. \quad (3)$$

The condition for a strong constraining potential is  $k \gg |E''|$ . To find the average values  $\langle y \rangle$  and  $\langle y^2 \rangle$ , we can write

$$\langle y \rangle = \frac{\int y e^{-\beta \mathcal{E}} dy}{\int e^{-\beta \mathcal{E}} dy}, \quad \langle y^2 \rangle = \frac{\int y^2 e^{-\beta \mathcal{E}} dy}{\int e^{-\beta \mathcal{E}} dy} \quad (4)$$

with

$$e^{-\beta \mathcal{E}} \sim e^{-\beta \{E'y + \frac{1}{2}(k + E'')y^2\}}, \quad (5)$$

where we have not included the constant term  $E(x_0)$ , and  $\beta \equiv 1/k_B T$  ( $k_B$ : Boltzmann constant,  $T$ : temperature). Since Eq. 5 is a Gaussian, moments of  $y$  are analytically integrable. After some algebra,

$$\langle y \rangle = \langle x - x_0 \rangle = -\frac{E'}{k + E''} \quad (6)$$

$$\langle y^2 \rangle - \langle y \rangle^2 = \text{var}(x - x_0) = \frac{k_B T}{k + E''}. \quad (7)$$

Generalizing this to three dimensions (3-D) is straightforward, and the  $i$ -th component of the force at  $\vec{r}_0$  is, by combining Eqs. 6 and 7,

$$F_i(\vec{r}_0) = -\frac{\partial}{\partial r_i} E|_{\vec{r}=\vec{r}_0} = \frac{k_B T}{\text{var}(r_i - r_{0i})} \langle r_i - r_{0i} \rangle. \quad (8)$$

Eq. 8 gives the ensemble averaged force applied at  $\vec{r}_0$  by the system. It does not explicitly involve the spring constant  $k$  of the sampling potential, reminiscent of the equipartition theorem (Reif, 1965). In the limit  $k \rightarrow \infty$ , Eq. 6 gives  $\vec{F} = k\langle \vec{r} - \vec{r}_0 \rangle$ , restoring Hooke's law.

## S.6 Simulation method for generating the force map

To implement the approach of Sec. S.5 in simulations, we used the region AA 1-12 (cover) and AA 317-331 (neck) from 1MKJ in the bound conformation of the crystal structure. As in the explicit water simulation of isolated CNBs, we fixed  $C_\alpha$  atoms in AA 9-12 and AA 317-323 at the base of CNB. The extra helical turn of  $\alpha 6$  in 1MKJ is formed between AA 321-325. Fixing AA 317-323 thus mimics conformation of  $\alpha 6$  in the ATP-bound state, yet allows some flexibility on the AA 324-325 to avoid a possible bias imposed by fixing. In 1MKJ, I325 is held by hydrophobic interactions with the switch II (Figure S8D), thus fixing the atoms at the base of CNB is expected not to create any extra conformational bias.

A 3-D version of the harmonic sampling potential (Eq. 1) was applied to V331  $C_\alpha$  with the center of the potential  $\vec{r}_0$  located at the bound position of V331  $C_\alpha$ , and with a spring constant  $k = 4.77 \text{ kcal}/(\text{mol} \cdot \text{\AA}^2)$  ( $\approx 3320 \text{ pN/nm}$ ). Under this potential, a freely moving atom fluctuates with an amplitude of  $0.5 \text{ \AA}$  at 300 K. Since V331  $C_\alpha$  can move up to several nm when the neck linker pointing backward moves forward, the corresponding energy surface should vary with a similar length scale. Since it is much larger than  $0.5 \text{ \AA}$ , the above choice for  $k$  satisfies the condition  $k \gg |E''|$ .

In addition, harmonic constraints on backbone hydrogen bonds were applied to CNB so that it does not break during MD. The first constraint was on the extra turn of  $\alpha 6$  that is formed only in the ATP-like conformation (Figure 6A). The turn actually ended as a  $3_{10}$  helix, so the constraints were on R321 O - T324 H and A322 O - I325 H, as identified in 1MKJ (the same arrangement was observed in 2KIN: R323 O - T326 H and A324 O - I327 H). To prevent disruption of the cover-neck  $\beta$ -sheet, additional harmonic constraints were applied to the backbone hydrogen bonds between A5-I325, C7-V329, and I9-N327. They had a spring constant of  $20 \text{ kcal}/(\text{mol} \cdot \text{\AA}^2)$  and were activated only when the distance between the atom pairs exceeded  $2.4 \text{ \AA}$ , which is a commonly used cutoff for hydrogen bonds (De Loof et al., 1992). We used these constraints to ensure the structural integrity of CNB during long sampling runs that scanned through points that included unstable high force

regions. In the more relevant low-force regions, the force generated by CNB is not significantly affected by these constraints (See Sec. S.9). Instability arises because the harmonic sampling potential forces V331  $C_\alpha$  to be at a sampling point, regardless of whether that point is physically accessible. Without the sampling potential, constraints were not necessary to keep CNB stable (Figures 2D-G and S13).

The sampling MD was performed in the ACE2 continuum solvent environment with the temperature maintained at 300, 330, or 360 K using the Nosé-Hoover thermostat. The simulation time was 440 ps at each sampling point. Coordinate trajectories were saved every 1 ps. The first 40 ps was used as an equilibration interval under the applied sampling potential, and average and variance in Eq. 8 were evaluated for V331  $C_\alpha$  during the last 400 ps. After sampling one point, a new sampling potential was applied whose center ( $\vec{r}_0$ ) is  $2 \text{ \AA}$  away from the previous one, and another 440 ps MD was performed. This procedure was repeated to cover the entire sampling space as shown in Figure S14. In total, there were 1626 points sampled at three different temperatures and at 300 K without the cover strand, which required about  $2.9 \mu\text{s}$  of simulation time.

Navigation within the sampling space was performed using the following procedure. First, a coordinate system was chosen so that the  $z$ -axis is along the axis of  $\alpha 6$ . The vector perpendicular to the  $z$ -axis and pointing to V331  $C_\alpha$  in its original bound conformation was then set as the  $-x$ -axis, making the  $+x$ -direction directly away from the neck binding pocket (Figure S14A). The  $y$ -axis is fixed by  $\hat{y} = \hat{z} \times \hat{x}$ . The coordinate origin was chosen at the end of  $\alpha 6$ . This choice of the coordinate system is for convenience in managing simulations, and has no effect on the result.

The initial set of sampling points were chosen on a plane that is perpendicular to the  $z$ -axis and that contains the position of V331  $C_\alpha$  in its bound conformation. Sampling points were shifted along concentric circles with radii decreasing by  $2 \text{ \AA}$  (Figure S14A). After the initial sampling slice was complete, the sampling points were sequentially varied along the  $z$ -axis by  $+2, +4, +6 \text{ \AA}$  or  $-2, -4, -6, -8 \text{ \AA}$  (Figure S14B). While further sampling in a larger space is possible, the spherical shell structure of the low-force region in Figure 5 suggests that the present sampling points cover typically reachable regions. For example, sampling points whose  $z$ -coordinates are more than  $+6 \text{ \AA}$  larger than the starting plane would cause the neck linker too stretched. Similarly, points whose  $z$ -components are shifted backward more than  $-8 \text{ \AA}$  would correspond to the situation in which the neck linker is under a large backward strain.

## S.7 Angular distribution of the generated force

Figure 5 qualitatively shows that the force generated by CNB is pointing to the direction of the neck binding

pocket, and that the directional bias disappears without the cover strand. We quantified this by measuring the angular distribution of forces. The best way to see this bias is to project the force on the  $xy$ -plane based on the coordinate system defined in Figure S14. The  $z$ -component of the force would be related more to the forward directionality, thus has a weaker relevance to the bias into the binding pocket.

In Figure S15A, we define the angular deviation  $\Delta\phi$  of the  $xy$ -projected force from that pointing directly into the binding pocket ( $\overrightarrow{PB}$ ). The sign of  $\Delta\phi$  is defined in a counterclockwise sense, so that a positive value means that the force vector is pointing the upper side of point  $B$ . The distribution of  $\Delta\phi$  thus represents the degree of conformational bias. As a control, we also calculated the distribution under the assumption that at every sampling point the force is pointing to the axis of the  $\alpha 6$  helix, not the binding pocket. Figure S15B shows that the distribution for CNB is sharply distributed near  $\Delta\phi = 0$ , demonstrating the conformational bias. Furthermore, it is nearly invariant at three different temperatures tested, indicative of the enthalpic nature of the generated force. The peak is shifted slightly to the positive value, so that the force vectors are directed to a point slightly higher than  $B$  in Figure S15A. On the other hand, the case without the cover strand has nearly a concentric distribution with no bias.

## S.8 Estimation of the stall force

Based on our results, we can relate the stall force  $F_s$  measured in optical trap experiments to the force required to prevent bending and subsequent docking of CNB. This is not simply the global minimum in the force map, since CNB could follow different paths along which higher forces are exerted; e.g., if an external force equal to the global minimum (28.2 pN; Figure 5) were used to oppose the docking of CNB, it could still dock through paths that require a higher stall force. While a detailed analysis of all possible paths would be prohibitive, we can use the calculated results to obtain an estimate for  $F_s$ . There are 361 sampling points with forces less than 150 pN (peak of the distribution in Figure 4). We choose  $F_s$  as the one below which the corresponding sampling points contribute 10% of this population (36 points); the resulting value is 64 pN. From the form of the distribution,  $F_s = 150$  pN would be an upper limit, so a value between 60~150 pN is suggested for the stall force.

The experimental value of 5-7 pN for the stall force (Block et al., 2003) cannot be compared directly with the calculated results: Most importantly, the role of the moment arm has to be considered (Figure S16). In laser trap experiments, the force applied through microbeads attached to the 80-nm long coiled-coil stalk has a moment arm that depends on the locations of hinges in the structural elements between the point of application of the force and  $\beta 9$ . The first hinge (Hinge 1 in Figure

S16) is on the neck stalk where the heptad repeat of the coiled-coil is disrupted (AA 564-606 for *Drosophila* kinesin, (Yang et al., 1989)). The resulting moment arm would be about 4 nm for 12AA neck linker plus 20~35 nm of a ~230 AA  $\alpha$ -helix (from the end of  $\beta 10$  to the hinge; 20 nm is obtained considering the coiled-coil geometry, as in the tropomyosin structure (PDB ID: 2B9C), while 35 nm is the length of a straight  $\alpha$ -helix). The second potential hinge (Hinge 2 in Figure S16) is located between  $\beta 10$  and the N-terminal end of the neck helix, which would be ~4 nm in length. In our simulation, the moment arm of V331 C $_{\alpha}$  is approximately 2.27 nm long. With  $F_s = 64$  pN applied at V331 C $_{\alpha}$ , the moment arm  $l_{exp}$  required to obtain  $F_{exp} \simeq 5-7$  pN would be  $2.27 \times 64 / (5 \sim 7) = 21 \sim 29$  nm long. This agrees well with the moment arm of Hinge 1. Supporting this is the result that in experiments in which the length of the coiled-coil stalk was varied, limping of kinesin disappeared in constructs whose lengths were beyond 600-870 AA (280-550 AA of coiled-coil stalk) (Asbury et al., 2003), which shows that Hinge 1 is flexible so that it could indeed determine the moment arm of the applied force. When the calculated peak force of 150 pN in Figure 5A is used,  $l_{exp} \simeq 49 \sim 68$  nm, which is still shorter than the ~80 nm length of the molecule, suggesting that the forces calculated from the simulations are in a range that is consistent with experimental measurements.

Alternatively, if the actual moment arm is determined by Hinge 2, the torque generated by CNB ( $\simeq 64 \times 2.9$  pN·nm) would exceed that by the laser trap ( $\simeq (5 \sim 7) \times 4$  pN·nm). This would mean that substeps other than CNB's walking stroke would be load-limiting. Possibilities are the formation of CNB or latching of  $\beta 10$ . In the first case, excessive rearward strain on the neck linker might lower its chance to form CNB, which potentially affects ATP binding as well (Rosenfeld et al., 2003). The latter is also a possibility since  $\beta 10$  could remain flexible after the forward motion of CNB. While existing data supports Hinge 1 as the actual point of action, further experiments are required to determine the load-limiting step. On the other hand, using the same criterion (a stall force of 13 pN as the 10% point of the population below 50 pN), forces generated by the neck linker alone yield a moment arm in the range 4.2~5.9 nm, about the length of the neck linker. This is either unrealistic, or if the neck linker in fact worked alone, its walking stroke must be load-limiting.

## S.9 Force map in explicit water

To test the reliability of the force map calculated from the implicit solvent simulation, we performed explicit water simulations on eight select points (Figure S17). Four pairs of nearby points, aligned approximately with the  $x$ - or  $y$ -axes in Figure S14A were selected on the same plane as that in Figure 5D. Within a pair, the same hydrogen bond constraints as those in the implicit solvent simulations were applied to the sampling run on one point

(orange spheres in Figure S17), and not on the other. This was to see the effect of the hydrogen bond constraint (Sec. S.6) on the generated force of CNB.

Procedures for solvation, heating, and equilibration followed steps similar to those in Sec. S.4. The production run lasted 400 ps. One sampling run took approximately 13 hours when eight Intel 3.0 GHz Xeon CPUs were used in parallel, compared to 37 minutes by one CPU for the corresponding implicit solvent simulation. Force map sampling procedure was similar to that in Sec. S.6, except that a stiffer spring constant of 10 kcal/(mol·Å<sup>2</sup>) was used for the sampling potential. We found that  $k = 4.77$  kcal/(mol·Å<sup>2</sup>) in explicit water was too soft and caused slow undulation of V331 C<sub>α</sub>, compromising proper sampling within the 400 ps run time. For  $k = 10$  kcal/(mol·Å<sup>2</sup>), a freely moving atom fluctuates by 0.34 Å at 300 K. Although in principle the limit  $k \rightarrow \infty$  would be ideal, in practice a very stiff sampling potential would render both the numerator and the denominator of Eq. 8 too small, leading to fluctuation errors.

The forces shown in Figure S17 are comparable to those obtained from implicit solvent simulations, both in terms of direction and magnitude. This was regardless of the presence or absence of the hydrogen bond constraints used to maintain the structural integrity of CNB, confirming that the major role of the constraint was to keep CNB from breaking in high force regions.

## S.10 Comparison with other experiments

It has been observed that the C330/L4 crosslinking (cf., Figure S8D) resulted in decrease in speed, run length, and stall force (Tomishige and Vale, 2000), consistent with the picture that formation/breakage of CNB is required for normal kinesin motility. The underlying assumption for the power stroke mechanism in Figure 6A is that unwinding of the extra turn of  $\alpha 6$  by the motion of switch II (as observed in the ADP state) also occurs in the nucleotide-free state. It was found that undocked neck linkers point in similar directions in both ADP and nucleotide-free states (Rice et al., 1999), so that the extra turn of  $\alpha 6$  is likely to be unwound as well.

In another study (Rice et al., 2003), the V331A/N332A mutation in the neck linker decreased the speed and the stall force of kinesin, but did not completely inhibit the forward processive motility. Based on the proposed picture, the two point mutations likely have different effects. N332 is the latch, so N332A will decrease tight binding of the neck linker. Since V331, due to its high  $\beta$ -sheet propensity (Kim and Berg, 1993), could promote CNB formation by  $\beta 9$  (I325-V329) or even transiently reinforce CNB by forming additional  $\beta$ -sheet, V331A may weaken the power stroke. As discussed in Supplemental Text S.8 regarding the stall force estimation, if latching rather than power stroke is load-limiting, N332A may also decrease the stall force, while the effect of V331A would be less prominent, unless it severely alters CNB

formation. The effect of the V331A/N332A mutation on kinesin's microtubule binding affinity was not significant, consistent with our view that the function of CNB is decoupled from other parts of the molecule. It has been suggested that hydrophobic interactions with the binding pocket are responsible for the neck linker docking (Rice et al., 2003), but our results argue against this: In Figure S8, ILE at the beginning of  $\beta 9$  has a hydrophobic contact with switch II. While this might help the extra helical turn of  $\alpha 6$  to form, located at the base of the neck linker, it alone cannot generate the power stroke. Furthermore, bending of an isolated CNB clearly demonstrates its conformational bias without any interaction with the binding pocket (Figure S13).

In Rice et al. (2003), the favorable enthalpy change and the unfavorable entropy change upon the neck linker docking were found to be both large, about 50 kJ/mol, with the net free energy change ( $\Delta G$ ) only about 3 kJ/mol  $\approx 1.2 k_B T$  ( $T = 300$  K). Our results suggest that CNB docking mainly involves an enthalpic stabilization, while its formation from the disordered cover strand and the neck linker leads to a loss in entropy, resulting in a small overall  $\Delta G$  (Supplemental Text S.11). The small value of  $\Delta G$  was used as evidence that kinesin's forward directionality arises from a Brownian ratchet (Mather and Fox, 2006); i.e., the exponentiation of a small  $\Delta G$  can lead to a large difference in forward and backward stepping rates. However, external loads will also be exponentiated, which can result in unlikely predictions such as extension of the coiled-coil stalk by 5 nm (Mather and Fox, 2006). In the model proposed here, a Brownian ratchet or a diffusive motion is a component rather than the sole mechanism, of kinesin motility. While the neck linker docking may have a small diffusive search step, other substeps could exhibit more ratchet-like character. For example, the forward and backward steps have nearly identical enthalpy (probably due to the strong binding of the motor head to the microtubule), but differ in entropy by  $6k_B$  (Taniguchi et al., 2005), which is larger than  $\Delta G$  of the neck linker docking. We postpone further discussion of the partitioning of the free energy among substeps to the full analysis of the dimer motility on the microtubule, as an extension of our present work.

## S.11 Thermodynamics of CNB formation

As CNB is a  $\beta$ -sheet between two flexible strands, its folding kinetics is likely to be similar to that of  $\beta$ -hairpin folding, where detailed data are available. In the case of the folding of a 16-residue long hairpin from the Ig-binding domain of protein G (Dinner et al., 1999), the entropy and enthalpy changes upon folding of the  $\beta$ -hairpin are both large and similar ( $\sim 33$  kJ/mol), with the enthalpy gain larger than entropy loss only by 0.84 kJ/mol. Although the hairpin can in principle form 7 hydrogen bonds along the backbone, entropic cost sets the state with 5 hydrogen bonds as the free energy minimum,

which is the same size as CNB. We thus expect that CNB formation would exhibit a similar thermodynamic behavior. While 33 kJ/mol is somewhat lower than the measured values of enthalpy or entropy changes for the neck linker docking ( $\sim 50$  kJ/mol) (Rice et al., 2003), there are contributions from substeps other than CNB formation, such as folding of the extra turn of  $\alpha 6$ , bending of CNB (power stroke, which is mostly enthalpic), binding of the ASN latch, and  $\beta$ -sheet formation between  $\beta 10$  and  $\beta 7$ . It can thus be inferred that CNB formation has a major contribution to the overall thermodynamic parameters of the neck linker docking.

## S.12 Comparison between available kinesin structures

Although it is unclear whether 1MKJ or 2KIN accurately represent the ATP state, their MH conformations correspond to those obtained with other ATP analogues. In Supplemental Table S2, we compare 43 available PDB structures. Structures are marked 'Docked,' if the  $\beta 9$  part of NL is in a forward-pointing, bound conformation, even if the  $\beta 10$  part is invisible (*cf.*, Figure S12). To put  $\beta 10$  in place, additional stabilization might be necessary through MT binding or through coordination with the other motor head in a dimer. In docked structures, switch II is in a 'retracting' position as in Figure 6A (orange). Note that an ATP-like docked structure can be obtained even with ADP; in the absence of MT, NL is considered to be in equilibrium between docked and undocked states, whose choice depends on the crystallization condition (Sindelar et al., 2002). Among docked structures, CNB is considered to be formed if two or more residues in CS form backbone hydrogen bonds with NL. In the case of 1MKJ or 2KIN, there are three such residues (A5, C7, and I9; Table 1, Figure S8)), while in most other cases there are two. Note that a CNB can be observed even when the N-terminal end of CS is invisible (Figure S12). However, during the power stroke, more backbone hydrogen bonds could form towards the N-terminal end of CS, making CNB longer. For Kif1A (Kinesin-3) and Eg5 (Kinesin-5), the structural arrangements of CS, NL, and switch II between docked and undocked states are similar to those between 1MKJ and 1BG2 in the Kinesin-1 family (Figure S12). Out of 18 x-ray structures of Kinesin-1, 3, and 5 in the docked state, there are only five (1YRS, 2FKY, 2FL2, 2FL6, and 2G1Q) that do not form CNB as defined above: In these structures, two CS residues are visible, and one still forms a backbone hydrogen bond with NL, that corresponds to I9 in 1MKJ. However, they are all for Eg5, obtained in related studies by Hartman and co-workers (Cox et al., 2005, 2006; Fraley et al., 2006). Compared to the stably folded MH core, NL does not bind strongly to MH, and CNB is only a transiently formed structure for power stroke generation, which probably account for variations in CS and NL conformations among existing structures.

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| PDB ID      | Nucleotide                            | Neck State              | Organism (Type <sup>1</sup> ) | Res. <sup>2</sup> | Reference                   |
|-------------|---------------------------------------|-------------------------|-------------------------------|-------------------|-----------------------------|
| 2KIN        | ADP/SO <sub>4</sub>                   | Docked/CNB              | Rat (KHC, 1)                  | 1.90              | Sack et al. (1997)          |
| 3KIN        | ADP/SO <sub>4</sub>                   | Docked/CNB              | Rat (KHC, 1)                  | 3.10              | Kozielski et al. (1997)     |
| <b>1BG2</b> | ADP                                   | <b>Undocked</b>         | Human (KHC, 1)                | 1.80              | Kull et al. (1996)          |
| <b>1MKJ</b> | ADP/SO <sub>4</sub>                   | <b>Docked/CNB</b>       | Human (KHC, 1)                | 2.70              | Sindelar et al. (2002)      |
| 1GOJ        | ADP                                   | Docked/CNB              | <i>N. crassa</i> (KHC, 1)     | 2.30              | Song et al. (2001)          |
| <b>1I5S</b> | ADP                                   | <b>Undocked</b>         | Mouse (Kif1A, 3)              | 2.20              | Kikkawa et al. (2001)       |
| <b>1I6I</b> | AMPPCP <sup>3</sup>                   | <b>Docked/CNB</b>       | Mouse (Kif1A, 3)              | 2.00              | Kikkawa et al. (2001)       |
| 1IA0        | AMPPCP                                | Docked/CNB              | Pig (Kif1A, 3)                | 15.00             | Kikkawa et al. (2000)       |
| <b>1VFV</b> | AMPPNP <sup>4</sup>                   | <b>Docked/CNB</b>       | Mouse (Kif1A, 3)              | 1.85              | Nitta et al. (2004)         |
| 1VFW        | AMPPNP                                | Docked/CNB              | Mouse (Kif1A, 3)              | 2.30              | Nitta et al. (2004)         |
| 1VFX        | ADP-Alfx <sup>5</sup>                 | Docked/CNB              | Mouse (Kif1A, 3)              | 2.55              | Nitta et al. (2004)         |
| <b>1VFZ</b> | ADP-Vanadate <sup>6</sup>             | <b>Undocked</b>         | Mouse (Kif1A, 3)              | 2.24              | Nitta et al. (2004)         |
| 2HXF        | AMPPNP                                | Docked/CNB              | Mouse (Kif1A, 3)              | 10.00             | Kikkawa and Hirokawa (2006) |
| 2HXH        | ADP                                   | Undocked                | Mouse (Kif1A, 3)              | 11.00             | Kikkawa and Hirokawa (2006) |
| <b>1I16</b> | ADP                                   | <b>Undocked</b>         | Human (Eg5, 5)                | 2.10              | Turner et al. (2001)        |
| <b>1Q0B</b> | ADP/Monastrol                         | <b>Docked/CNB</b>       | Human (Eg5, 5)                | 1.90              | Yan et al. (2004)           |
| 1YRS        | ADP/L47 <sup>7</sup>                  | Docked                  | Human (Eg5, 5)                | 2.50              | Cox et al. (2005)           |
| 1X88        | ADP/Monastrol                         | Docked/CNB              | Human (Eg5, 5)                | 1.80              | (Maliga, unpublished)       |
| 2IEH        | ADP/Monastrol                         | Docked/CNB              | Human (Eg5, 5)                | 2.70              | (Garcia-Saez, unpublished)  |
| 2FKY        | ADP/Inhibitor <sup>8</sup>            | Docked                  | Human (Eg5, 5)                | 2.30              | Fraley et al. (2006)        |
| 2FL2        | ADP/Inhibitor                         | Docked                  | Human (Eg5, 5)                | 2.50              | Fraley et al. (2006)        |
| 2FL6        | ADP/Inhibitor                         | Docked                  | Human (Eg5, 5)                | 2.50              | Fraley et al. (2006)        |
| 2FME        | ADP/Inhibitor                         | Docked/CNB              | Human (Eg5, 5)                | 2.10              | Tarby et al. (2006)         |
| 2G1Q        | ADP/Inhibitor                         | Docked                  | Human (Eg5, 5)                | 2.51              | Cox et al. (2006)           |
| 2GM1        | ADP/Inhibitor                         | Docked/CNB              | Human (Eg5, 5)                | 2.30              | Kim et al. (2006)           |
| 1T5C        | ADP/NO <sub>3</sub> /PIN <sup>9</sup> | Docked                  | Human (CENP-E, 7)             | 2.50              | Garcia-Saez et al. (2004)   |
| 2NR8        | ADP                                   | Absent in the construct | Human (Kif9, 9)               | 2.00              | (Zhu, unpublished)          |
| 1V8J        | ADP                                   | N/A                     | Mouse (Kif2C, 13)             | 3.24              | Ogawa et al. (2004)         |
| 1V8K        | AMPPNP                                | N/A                     | Mouse (Kif2C, 13)             | 2.25              | Ogawa et al. (2004)         |
| 1RY6        | None (SO <sub>4</sub> )               | N/A                     | <i>P. falciparum</i> (13)     | 1.60              | Shiple et al. (2004)        |
| 2GRY        | ADP                                   | N/A                     | Human (Kif2, 13)              | 2.35              | (Wang, unpublished)         |
| 2HEH        | ADP                                   | N/A                     | Human (Kif2C, 13)             | 2.15              | (Wang, unpublished)         |
| 2NCD        | ADP                                   | N/A                     | <i>Drosophila</i> (Ncd, 14)   | 2.50              | Sablin et al. (1998)        |
| 1CZ7        | ADP                                   | N/A                     | <i>Drosophila</i> (Ncd, 14)   | 2.90              | Kozielski et al. (1999)     |
| 1N6M        | ADP                                   | N/A                     | <i>Drosophila</i> (Ncd, 14)   | 2.50              | Yun et al. (2003)           |
| 1SDM        | ADP                                   | N/A (Docked)            | Potato (KCBP, 14)             | 2.30              | Vinogradova et al. (2004)   |
| 1F9T        | ADP                                   | N/A                     | Yeast (Kar3, 14)              | 1.50              | Yun et al. (2001)           |
| 1F9U        | ADP                                   | N/A                     | Mutant of 1F9T (N650K)        | 1.70              | Yun et al. (2001)           |
| 1F9V        | ADP                                   | N/A                     | Mutant of 1F9T (R598A)        | 1.30              | Yun et al. (2001)           |
| 1F9W        | ADP                                   | N/A                     | Mutant of 1F9T (E631A)        | 2.50              | Yun et al. (2001)           |
| 3KAR        | ADP                                   | N/A                     | Yeast (Kar3, 14)              | 2.30              | A.M.Gulick et al. (1998)    |
| 2H58        | ADP                                   | N/A                     | Human (KifC3, 14)             | 1.85              | (Wang, unpublished)         |
| 2O0A        | N/A                                   | N/A                     | Yeast (Vik1, 14)              | 1.60              | Allingham et al. (2007)     |

Type<sup>1</sup>: Common names and/or standardized kinesin family number (Lawrence et al., 2004).

AMPPCP<sup>3</sup>: adenylyl 5'-( $\beta$ , $\gamma$ -methylene)diphosphonate (ATP analogue).

ADP-Alfx<sup>5</sup>: ADP-Alumino fluoride complex (ADP-Pi analogue).

L47<sup>7</sup>: 3,5-diaryl-4,5-dihydropyrazoles.

PIN<sup>9</sup>: 1,4-piperazinediethanesulfonic acid.

Res.<sup>2</sup>: Resolution in Å.

AMPPNP<sup>4</sup>: 5'-adenylylimidodiphosphate (ATP analogue).

ADP-Vanadate<sup>6</sup>: ADP-Pi analogue.

Inhibitor<sup>8</sup>: Various series of the mitotic kinesin inhibitors.

**Table S2:** Comparison between available kinesin structures. Those shown in Figures 6A, S12 are marked bold. Even with ADP, it is known that ATP-like neck-bound states can be achieved in the presence of other molecules, such as sulfate ions (Sindelar et al., 2002). For example, 2KIN, 1MKJ, and 1Q0B are in ATP-like states. Structures marked with 'Docked/CNB' have between 2-3 residues in the cover strand forming backbone hydrogen bonds with the neck linker. Transition state nucleotide analogues are used for 1VFX and 1VFZ. 1IA0, 2HXF and 2HXH are cryo-EM structures of the kinesin-microtubule complex. 2HXF and 2HXH respectively use 1VFV and 1I5S to fit the electron density map. In 1YRS, 2FKY, 2FL2, 2FL6, and 2G1Q, most of the cover strand is invisible, and only one residue at its base forms a backbone hydrogen bond with the neck linker. However, they are from the same group in similar studies, thus are possibly obtained under similar crystallization conditions. In 1T5C,  $\beta$ 10 forms a  $\beta$ -sheet with the motor head core, while  $\beta$ 9 does not have a secondary structure and no backbone hydrogen bond with the cover strand (invisible) is present. It is not clear whether this is a crystallization artifact or if the cover strand has less affinity with the neck linker in Kinesin-7. Kinesin-13 (microtubule depolymerizing) and Kinesin-14 (minus-end directed) do not have the neck linker, so their neck states are not accessed. In the case of 1SDM, it has a neck-linker mimicking domain which is docked in the structure. 2O0A does not have a nucleotide binding domain.



Figure S7: Sequence alignment of 2KIN (upper row) and 1MKJ (lower row). They have 86.2 % sequence identity and are structurally very similar, with less than 0.8 Å root mean square deviation between the corresponding C<sub>α</sub> atoms. Residue numbers immediately to the right of vertical dotted lines are shown on top (2KIN) and bottom (1MKJ) of each line. Secondary structures are colored as: α-helix (cyan), β-sheet (yellow), and loop (pink). In 2KIN, D293 is a cloning artifact and is originally G293 (Sack et al., 1997). 1MKJ is shorter than 2KIN by 2 residues at positions in AA 42-43 and AA 272-273 (marked by dashes). Although the switch II cluster is designated to a region spanning L11 to L13 (AA 232-296 in 2KIN) (Kikkawa et al., 2001), we use the residue numbers from Sindelar et al. (2002) to focus on its conformational change relevant to the neck linker docking. L11 is invisible in 1MKJ and 2KIN. There are no clear secondary structures before and after L11. The ASN latch is circled in blue.

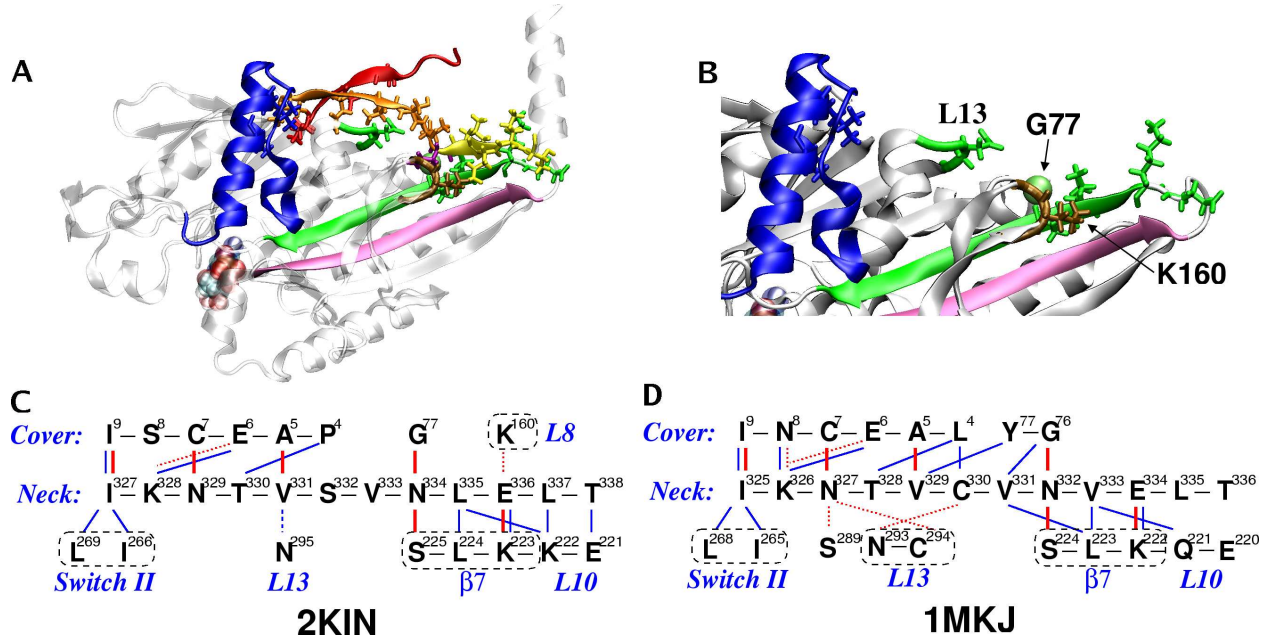


Figure S8: Contact map of the neck linker. (A) Overview of the residues of  $\beta 7$ ,  $\beta 9$ ,  $\beta 10$ , L8, L13, and the cover strand in contact in 2KIN (licorice representation). Viewing direction is the same as in Figure 1A. (B) Magnified view of the neck binding pocket in 2KIN. For clarity, CNB is not shown and G77 is represented as a sphere. (C,D) Interaction maps of 2KIN (C) and 1MKJ (D), with a distance cutoff of 2.4 Å. Thick red: backbone hydrogen bonds. Dotted red: salt bridge between polar side chains, or hydrogen bond between polar side chain and backbone hydrogen or oxygen. Blue: hydrophobic contact between side chains. Dotted blue: contact between V331 C $\alpha$  and the polar amino group of N295, which is likely to be weak. While 2KIN has higher resolution than 1MKJ (Supplemental Table S2), contacts with L8 and L13 in 2KIN are either absent or different in 1MKJ, suggesting that they may not be involved in specific binding with the neck linker compared to the cover strand or  $\beta 7$ . In 1MKJ, also note that L10 does not interact with  $\beta 10$  (L335-T336). N334 in 2KIN (N332 in 1MKJ) has two backbone hydrogen bonds (to G77 and S225), supporting its role as a latch. No domain name is assigned for the loop containing G77(G76) (Figure S7). We analyzed the crystal water molecules and found no water-mediated bridges between the neck linker and the binding pocket.

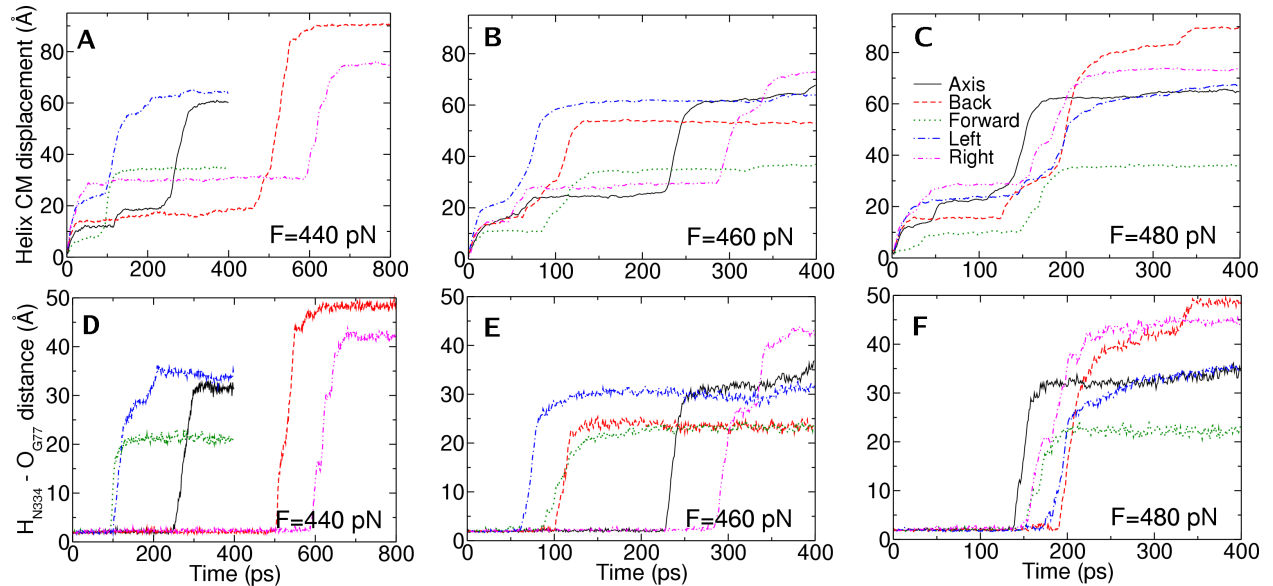


Figure S9: Trajectories in the neck pulling simulations of 2KIN (cf., Figure 1B,C). (A-C) Center of mass displacement trajectory of the pulled helix. (D-F) Distance trajectory between N334 H and G77 O, exhibiting a sharper transition than the center of mass trajectory. This shows that there is no strong affinity of  $\beta 9$  to the motor head after the release of the ASN latch.

Figure S10: Motion of the unbound head of 1MKJ (*cf.* Figure 2B,C). The fluctuation in the case without the cover strand is attenuated due to non-specific binding of the neck linker on the motor head (*cf.*, Figure S11B vs. C).

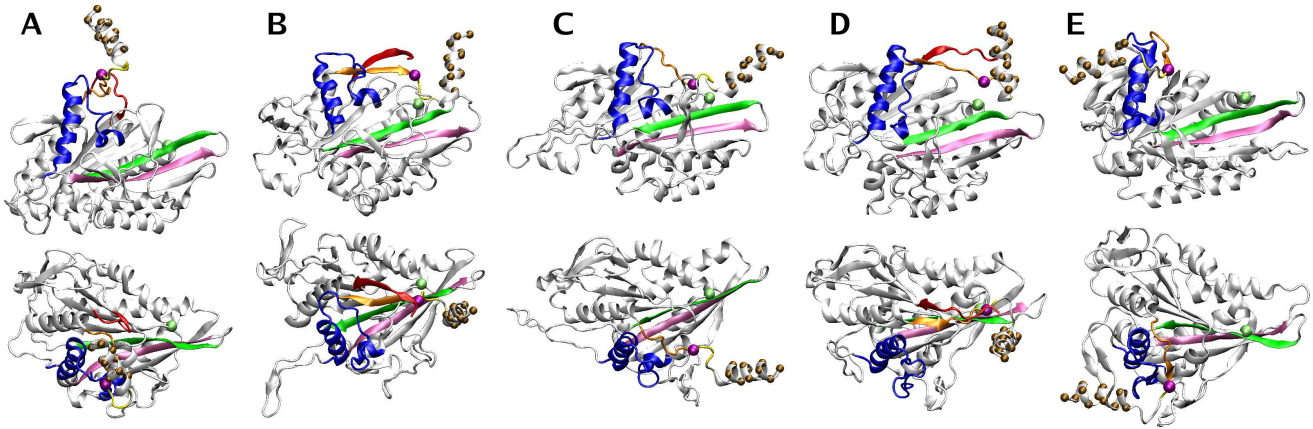
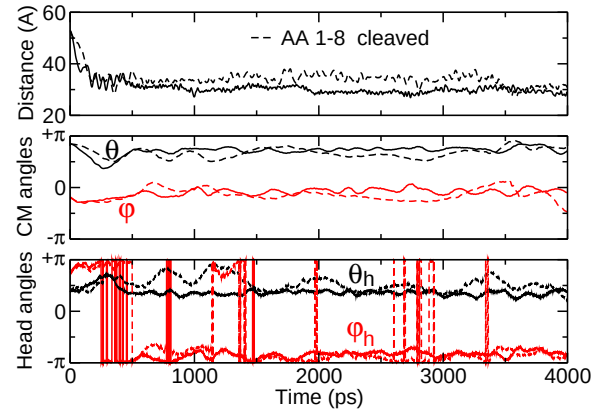


Figure S11: Final configurations of freely moving motor heads (*cf.* Figure 2). Top row:  $\beta 6$  is oriented in a direction similar to that in Figure 1A. Bottom row: a view similar to Figure 1B. Color codings are the same as in Figure 1A. The harmonically constrained  $C_{\alpha}$  atoms in the neck helix are denoted by brown spheres. (A) 1BG2 (8 ns, 5.5 Å; simulation time and the root mean square deviation (rmsd) of the  $C_{\alpha}$  coordinates of the motor head core (AA 10-323) from the starting structure), (B) 1MKJ (4 ns, 4.0 Å) (C) 1MKJ, with the cover strand cleaved (4 ns, 4.9 Å), (D) 2KIN (4 ns, 4.3 Å) (E) 2KIN, with the cover strand cleaved (4 ns, 5.1 Å). The rmsd indicates large conformational fluctuations of the motor head at the high simulation temperature (360 K). Nevertheless, comparing Figure 1A and (B,D) shows that CNB is in a conformation similar to the original bound state. In (A,C,E), the neck helix or the flexible neck linker binds non-specifically to the motor head, attenuating its motion (Figures 2C,S10).

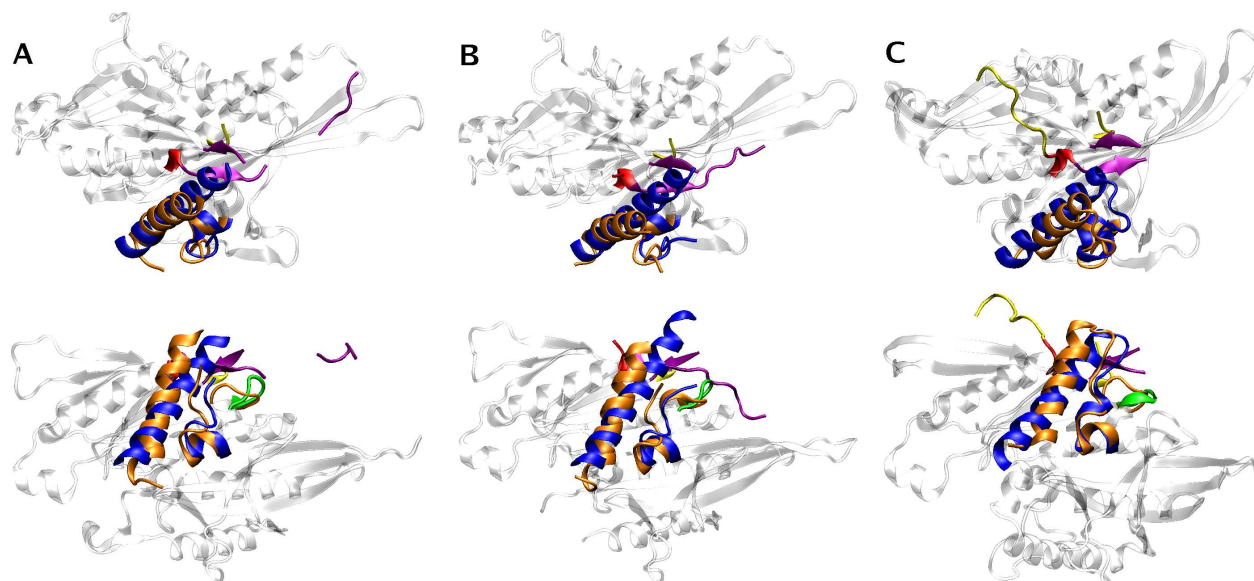


Figure S12: Other PDB structures with partially visible CNB (*cf.*, Supplemental Table S2). (A) 1I6I/1I5S (Kif1A), (B) 1VFV/1VFZ (Kif1A), (C) 1Q0B/1II6 (Eg5). Viewing directions are similar to those in Figure 6A (top row) and Figure 1A (bottom row). Color codes are: CNB (purple), cover strand and neck linker in unbound states (yellow), C-terminal end of  $\alpha 6$  that forms an extra helical turn in the neck bound state (red), switch II in the unbound (blue) and bound (orange) states, and L13 in the unbound (green) and bound (orange) states (*cf.* Figures 1A and 2F,G). L13 is highlighted only in the bottom row. In (A), the end of  $\beta 10$  in the bound state is partially visible, while it is invisible in (C). Thus the 'bound state' here means the conformation in which the switch II retracted to allow the extra turn of  $\alpha 6$  and subsequently CNB to form, not a state in which the neck linker is completely docked as in 1MKJ or 2KIN. In (A,B), L13 is in an upward protruding conformation in the unbound state, similar to Figure 2F,G. In (C), it protrudes forward rather than upward in the unbound state, which can also interfere with the neck linker binding.

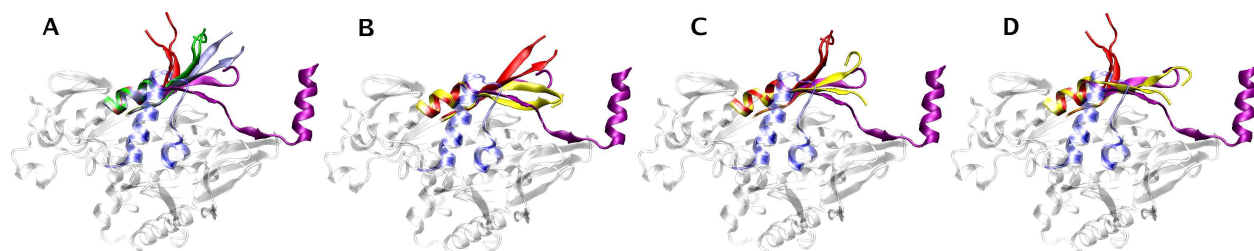


Figure S13: Explicit water simulation of isolated CNBs. The 1MKJ structure is superposed as a reference. Viewing directions are the same as in Figure 1A. The bound CNB in 1MKJ is in purple. (A) Superposition of three initial conformations for comparison. The one in pale blue is from the neck pulling simulation, which is also in (B). The other two are from the force map simulations. (B-D), Initial (red) and final (yellow) conformations. The final conformations are after (B,C) 0.6 ns, and (D) 1 ns. Even though isolated CNBs are used, its conformation after MD is similar to the bound conformation in 1MKJ. In (D), note that CNB can generate forward motion without breaking even when it is bent backward by more than  $90^\circ$ .

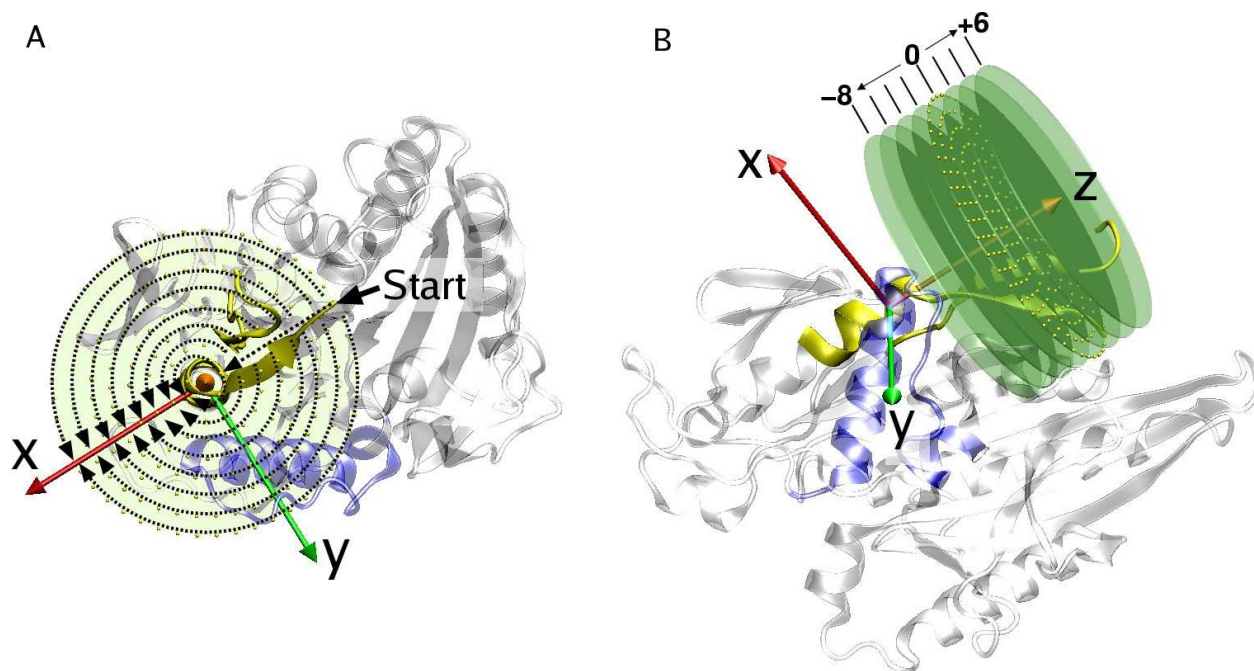


Figure S14: Sampling paths for the force map simulation. (A) Path on the initial slice (marked '0' in (B)). Starting from the position of V331  $C_{\alpha}$  in the bound conformation, the center  $\vec{r}_0$  (cf. Eq. 8) of the sampling potential independently followed the paths shown by dotted arrows from the outer circle. At each point the sampling time was 440 ps. The same view as in Figure 5D. (B)  $z$ -components of individual sampling points in the initial sampling slice were varied as indicated. Viewing direction is the same as in Figure 5A.

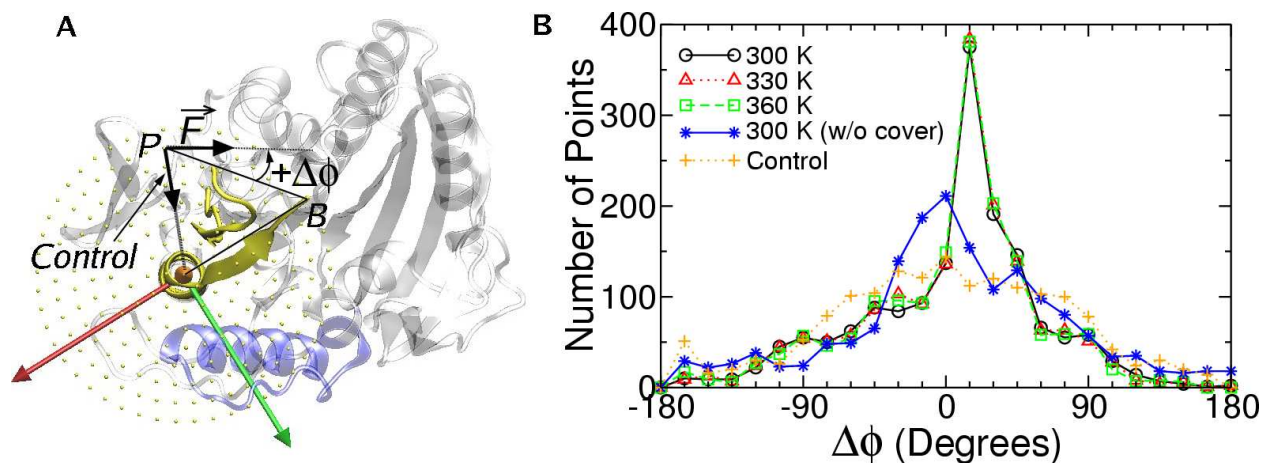


Figure S15: Angular distribution of force vectors in the azimuthal direction (cf., Sec. S.7). (A) Deviation  $\Delta\phi$  of the direction of the force projected on the  $xy$ -plane from the ideal bias along  $\vec{P}\vec{B}$ . Control: perfectly concentric distribution of force vectors.  $P$ : sampling point,  $B$ : position of V331  $C_{\alpha}$  in the bound conformation. (B) Comparison between different cases. Note that the case without the cover strand is closer to the perfect concentric case.

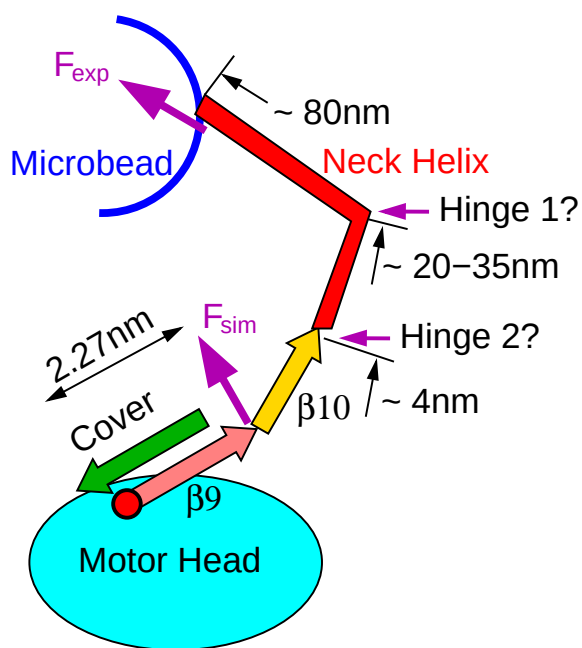


Figure S16: Estimation of the stall force as the minimum force required to prevent docking of the neck linker. Parts of the molecule are not drawn to scale. The N-terminal base of the neck linker is marked by a red circle as a hinge point from which the moment arm is measured. The stall force calculated from the simulation,  $F_{sim}$ , is exerted at the end of  $\beta 9$ , whose moment arm is  $l_{sim} \sim 2.27$  nm long. In laser trap experiments, the measured stall force  $F_{exp} = 5 \sim 7$  pN is applied through the microbead attached to the neck coiled-coil stalk. There are two possible flexible hinge points, denoted by *Hinge 1* and *Hinge 2*. Hinge 1 is located where the heptad repeat of the neck helix is interrupted. Hinge 2 is along or at the end of  $\beta 10$ , which may be flexible since it is a single peptide strand when the neck linker is not bound to the motor head. However, AA 1-4 of the cover strand (AA 5-9 participates in the CNB formation) might interact with  $\beta 10$  or with the base of the neck helix, stiffening this region even when the neck linker is unbound. Experiment suggests that Hinge 1 is more likely (Asbury et al., 2003). The figure depicts the case when Hinge 1 is flexible while Hinge 2 is not. In both experiment and simulation, torques generated by the corresponding stall forces should be the same, as they oppose the action of CNB. Denoting the moment arm in the experiment by  $l_{exp}$ , the following relation should hold:  $F_{sim} \times l_{sim} \simeq F_{exp} \times l_{exp}$ , which was used to estimate  $l_{exp}$  in Sec. S.8.

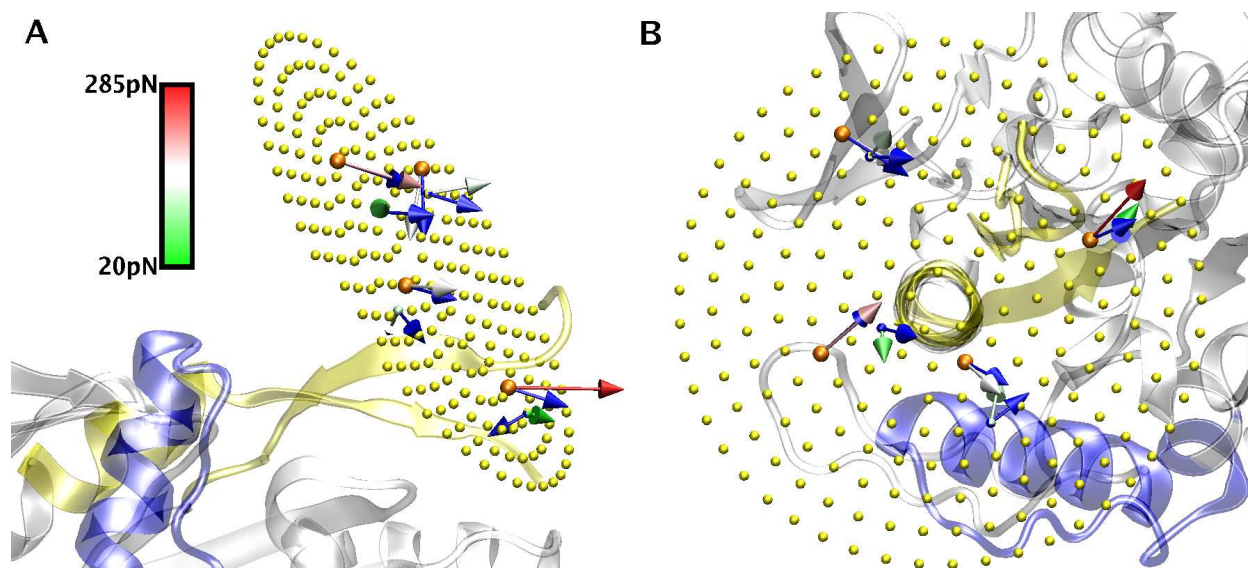


Figure S17: Force vectors obtained from explicit water simulations. Four points in which the hydrogen bond constraints were applied as in the implicit solvent simulation are denoted as orange spheres. For comparison, force vectors from implicit solvent simulations are shown in blue. Forces from explicit water simulation range between 24.5-283 pN, while those from the implicit solvent simulation range between 66.3-178 pN. The forces overall point to similar directions. Viewing directions are the same as Figure 5A (A), and Figure 5C (B). See Sec. S.9.